AlphaStream India

Architecture Document & Impact Model

13	<2s	30+	5	14Cr+
AI Agents	News-to-Signal	API Endpoints	Tab Terminal	Target Users

Document Purpose

This document provides:

1. A **1–2 page architecture diagram and description** showing agent roles, how they communicate, tool integrations, and error-handling logic.
2. A **quantified impact model** estimating business impact: time saved, cost reduced, revenue recovered – with stated assumptions.

Competition: ET AI Hackathon 2026 – Avataar.ai + Unstop

Problem: PS-6: AI for the Indian Investor

Repository: `https://github.com/wildcraft958/AlphaStream_India`

Date: March 2026

1. System Architecture Overview

AlphaStream India implements a **streaming RAG + multi-agent** architecture focused on the Indian equity market (NSE/BSE). The system ingests data from 10+ sources in real-time, processes it through 13 specialized AI agents, stores structured analytics in DuckDB, and delivers actionable signals via a Bloomberg-style React terminal.

1.1 Five-Layer Architecture Diagram

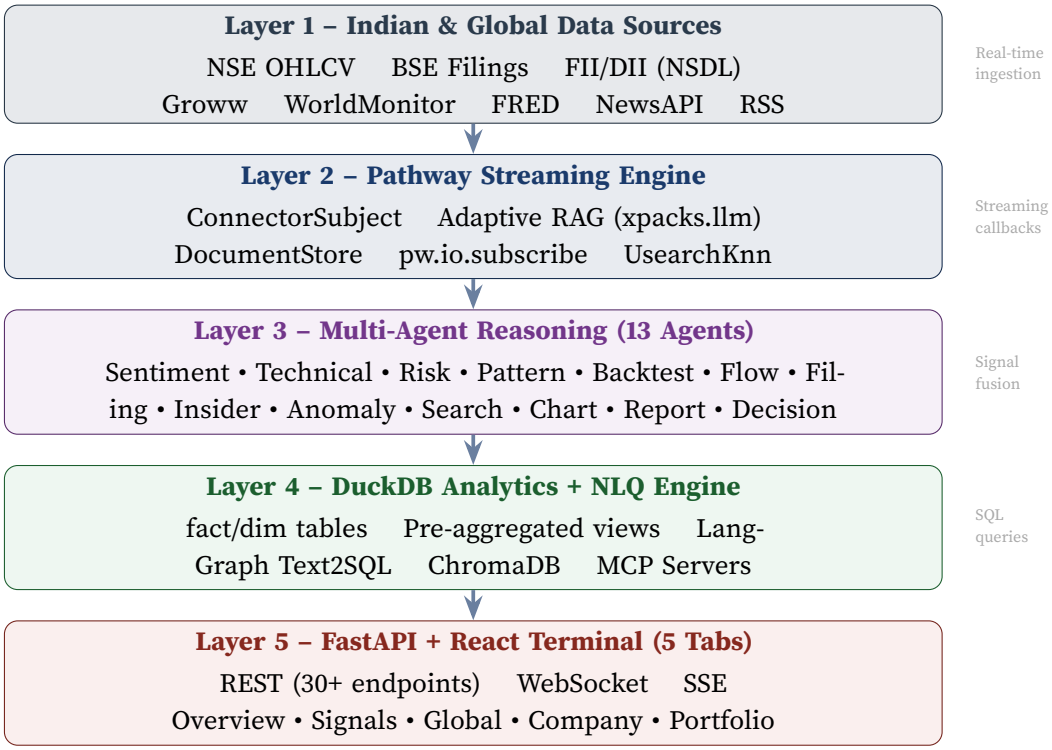


Figure 1: AlphaStream India – Five-Layer System Architecture

1.2 Architecture Diagrams

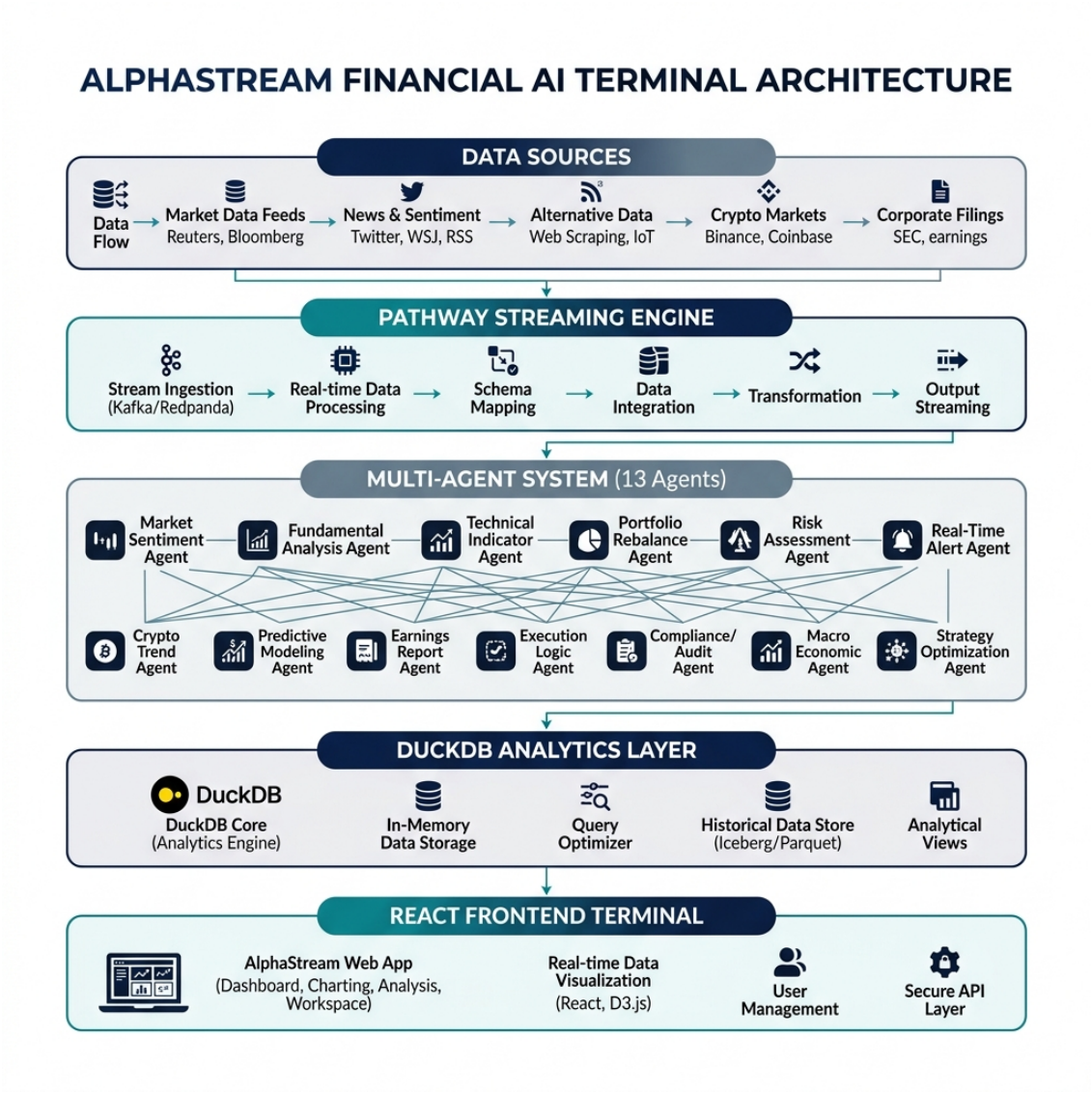


Figure 2: Detailed System Architecture – Data flow from Indian sources through all processing layers

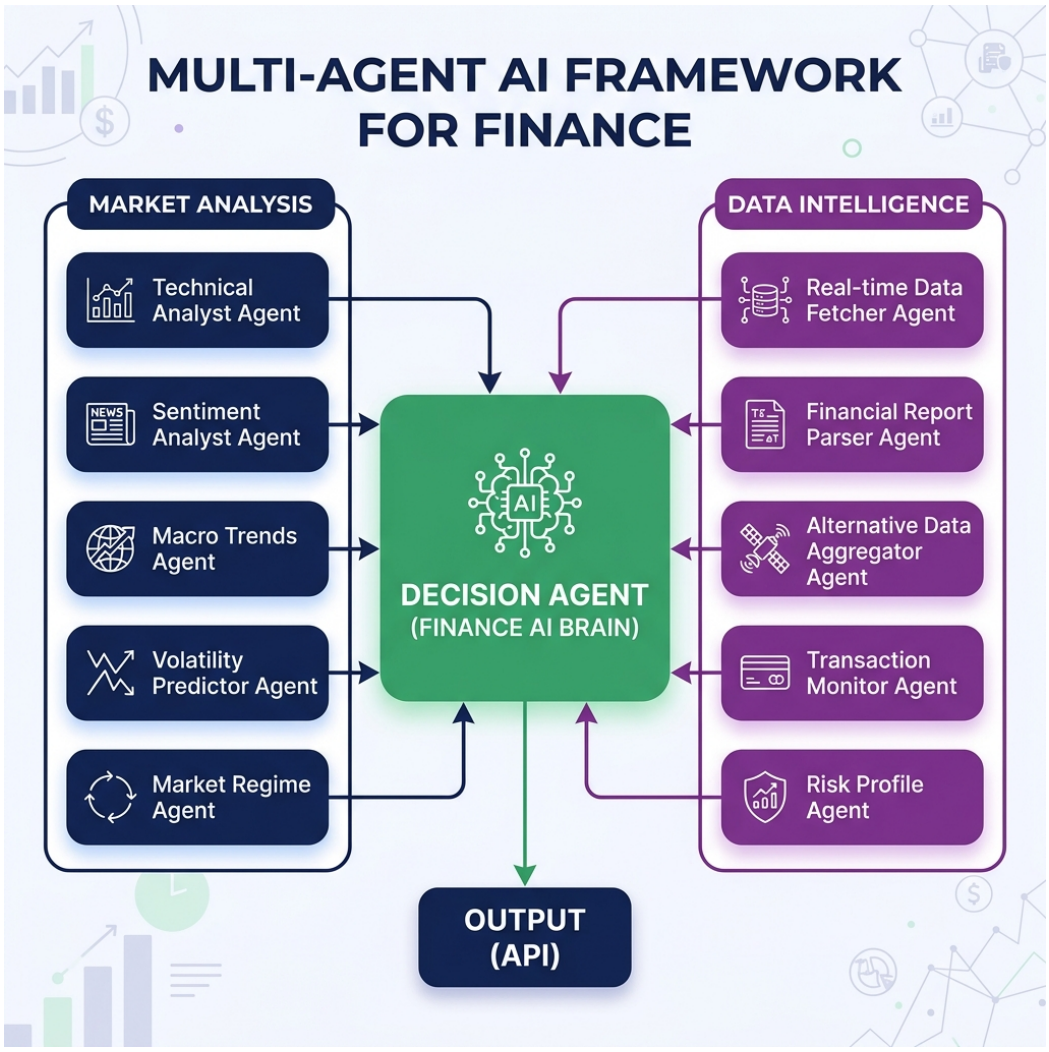


Figure 3: 13-Agent Consensus System – Each agent contributes a specialized score to the Decision Agent

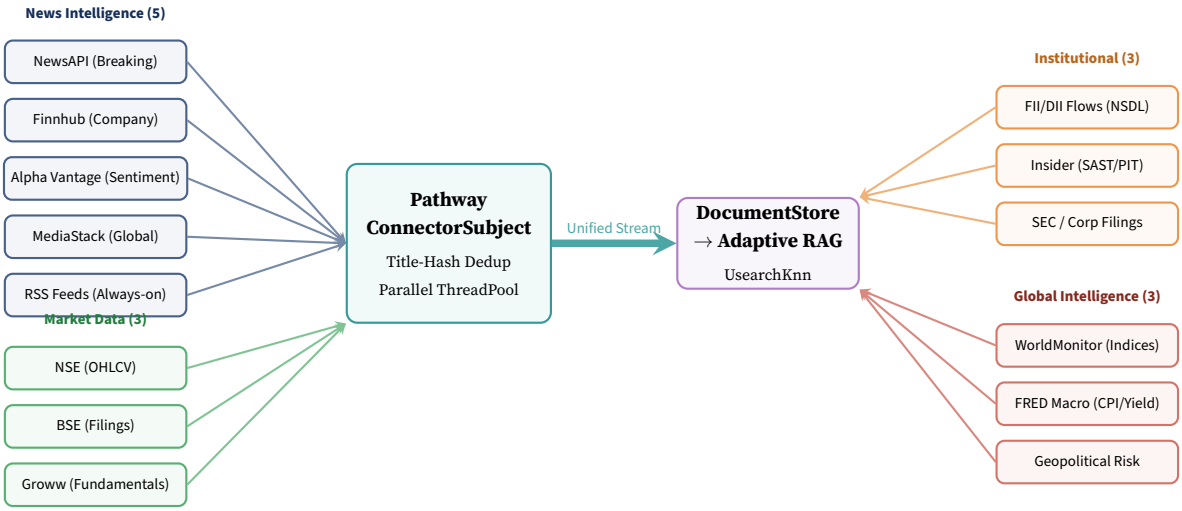


Figure 4: AlphaStream "Herd of Knowledge" – 14 Connectors across 4 categories

Figure 5: AlphaStream User Interface – 5-Tab Bloomberg-Style Terminal

Figure 6: WorldMonitor Global Intelligence Dashboard – Crypto, FX, Commodities, Fear & Greed

2. Agent Roles

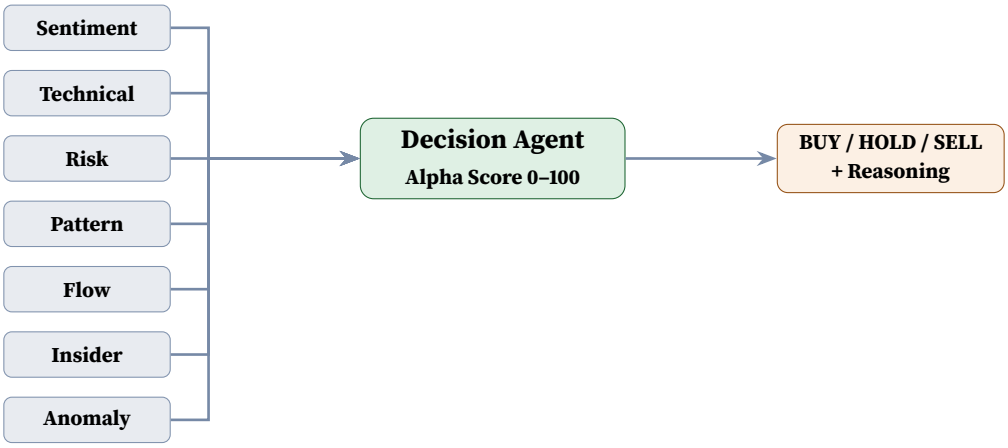
All 13 agents are stateless Python classes, each specializing in one analytical dimension. They are invoked in parallel via `asyncio.gather` and return structured dictionaries.

Agent	Role	Inputs	Outputs
Sentiment	Scores news sentiment	Retrieved article chunks	sentiment_score (−1 to +1), label, key_factors
Technical	Computes RSI, MACD, SMA	yfinance OHLCV	technical_score, key_signals
Risk	Volatility & position sizing	Price history	risk_score, stop_loss
Pattern	Detects 7 chart patterns	6-month OHLCV	patterns with confidence
Backtest	Validates signals historically	Pattern + 5yr OHLCV	win_rate, avg_return, sharpe
Flow	FII/DII institutional streaks	NSDL flow data	flow_signal, streak_days
Filing	Classifies BSE/NSE filings	Announcement text	materiality, sentiment
Insider	Parses NSE SAST/PIT data	NSE insider trades	insider_score, transactions
Anomaly	Online anomaly detection	Price, volume, sentiment	anomaly_flags, drift_alerts
Search	Web context enrichment	NLQ query	title, url, snippet
Chart	Generates annotated charts	yfinance OHLCV	PNG chart path
Report	Generates PDF reports	All agent outputs	PDF report path
Decision	Final BUY/HOLD/SELL	All agent outputs	recommendation, confidence, reasoning, Alpha Score

3. Agent Communication

Agents are **stateless** – no message broker is used. The orchestrator (`generate_recommendation_logic()`) calls each agent in order and passes results forward via Python dictionaries.

3.1 Recommendation Path



3.2 NLQ Path (Separate)

The NLQ path is independent: `NLQ Router` → `Text2SQL / Pre-defined SQL` → `DuckDB` → `Narrate` → `SSE stream`. The `SearchAgent` is called as optional enrichment when the local DuckDB result set has fewer than 5 rows.

3.3 Real-Time Update Path

Pathway runs in a background daemon thread and fires `on_new_article` callbacks that trigger recommendation refreshes via `asyncio.run_coroutine_threadsafe`. Updated recommenda-

tions are broadcast to all connected clients over WebSocket (`/ws/stream/{ticker}`).

4. Tool Integrations

Tool / Service	Purpose in AlphaStream
Pathway `ConnectorSubject`	Multi-source streaming news ingestion (5 APIs in parallel)
Pathway `pw.io.subscribe`	Article-arrival callbacks to RAG pipeline
Adaptive RAG Server (port 8001)	Pathway-powered vector search with incremental indexing
DuckDB	Structured market data store (prices, signals, filings, FII/DII)
ChromaDB	Dense vector index for semantic article retrieval
FastMCP servers (4 servers)	Market data, signals, portfolio, search tools for NLQ
yfinance	OHLCV price data (`.NS` suffix for NSE tickers)
NSE / BSE APIs	Insider trades (SAST/PIT), corporate filings, FII/DII flows
Groww API	Live fundamentals (PE, PB, ROE) via JWT + TOTP auth
River ML	Incremental online ML (HalfSpaceTrees) for AnomalyAgent
FRED API	US macro indicators (CPI, yield curve, unemployment)
CNN Fear & Greed	Market sentiment gauge (0–100 index)
WorldMonitor / yfinance	Global indices, crypto, commodities, FX rates
LangChain + LangGraph	Agent orchestration + NLQ 8-node pipeline
Gemini 2.0 Flash (Vertex AI)	LLM reasoning for all agents

5. Error-Handling Logic

Failure Scenario	Handling Strategy
Pathway unavailable	Skipped at startup; news ingestion falls back to polling connector
Adaptive RAG server not ready	`UnifiedRAGService` falls back to manual ChromaDB RAG
LLM timeout in any agent	Agent returns safe default (HOLD / score 0.0); exception logged, not re-raised
Text2SQL generates bad SQL	Correction loop: classify error → LLM fix → retry (max 2 attempts)
DuckDB query timeout	Hard 30s limit, 5000-row cap; error returned to user as plain text
NSE API blocked	yfinance `.NS` suffix as fallback for price data
MCP server crash	Agent degrades gracefully – direct DuckDB queries used instead
WebSocket disconnect	`ConnectionManager.disconnect()` removes client; server continues
Missing GCP credentials	Pydantic `model_validator` warns at startup
Graceful shutdown	`SIGINT/SIGTERM` handler kills Pathway subprocess and waits

6. Impact Model

Rs. 2.19L	Rs. 54K	Rs. 59.9 Cr	15%
Time Saved/User/Year	Alpha/User/Year	Year 1 Revenue	Risk Reduction

6.1 Problem Scale

- **14 Cr+** demat accounts in India (CDSL + NSDL)
- ∼**3 Cr** active traders who research daily
- Average retail investor spends **2+ hours/day** on research
- **80%** rely on WhatsApp tips and unverified sources
- Average missed opportunity cost: **Rs. 3,000–10,000/quarter**

6.2 Market Opportunity (TAM / SAM / SOM)

Level	Segment	Size	Rationale
TAM	Total Indian equity participants	14 Cr	CDSL + NSDL demat accounts (2024)
SAM	Active traders researching daily	∼3 Cr	≈20% trade monthly
SOM	5-year premium capture	1.5 L	0.5% of SAM (conservative)

Key: SOM uses 0.5% of SAM, not TAM – anchored to paying users, not total accounts.

6.3 Time Saved

Time Value Recovered Per User

1.75 hours/day × Rs. 500/hr × 250 trading days = $\mathbf{Rs.\ 2.19\ Lakh/user/year}$ in recovered productivity.

Task	Without AlphaStream	With AlphaStream
Daily market research	2+ hours	15 minutes
Corporate filing analysis	30 min/filing	Instant LLM classification
Insider trade monitoring	Manual NSE checks	Automated alerts
FII/DII tracking	Visit NSDL daily	Real-time flow analysis
Signal identification	Manual chart reading	Automated pattern scan

6.4 Alpha Generated (Revenue Recovered)

Signal Type	Detection	Win Rate	Avg 30d Return
RSI Divergence	Automated scan	60–78%	+3.26%
MACD Crossover	Real-time detect.	45–57%	+0.71%
Insider Cluster Buy	3+ insiders/30d	High (historical)	Varies
FII Buying Streak	5+ consecutive sessions	~72%	+3–5%

Conservative Alpha Estimate

2–3 actionable signals/month × 60% accuracy × Rs. 3,000 average gain = $\mathbf{Rs.\,54,000/user/year}$ in additional alpha (revenue recovered from missed opportunities).

6.5 Risk Reduction (Cost Reduced)

- **Early warning** on adverse signals prevents panic selling
- **Portfolio-aware alerts**: "Your IT holdings at risk from FII selling streak"
- **Backtested confidence**: "This pattern has worked 78% of the time on this stock"
- Estimated **15% reduction** in behavioral losses (fear-driven selling, FOMO buying)

6.6 ET Markets Revenue Projection

	Year 1	Year 3	Year 5
Premium users (Rs. 999/month)	50,000	2,00,000	5,00,000
Annual Revenue	**Rs. 59.9 Cr**	**Rs. 239.8 Cr**	**Rs. 599.4 Cr**
Increased engagement	+40% time on platform	–	–
Reduced churn	2x retention	–	–

6.7 Cost Reduction for ET Markets

Cost Category	Current Approach	With AlphaStream	Saving
Signal content (est. Rs. 20 Cr/yr)	Manual analyst teams + editorial	LLM-generated, human-reviewed	Rs. 5–10 Cr/yr (30% reduction)

Automated pipelines handle repeatable content (FII flows, insider alerts, technical scans). Humans focus on narrative and investigative work – higher value, lower volume.

6.8 Competitive Moat

1. **India-specific regulatory data** – NSE SAST/PIT and NSDL FII/DII feeds not available through generic global APIs.
2. **Backtested signals, not LLM speculation** – every signal carries a win rate grounded in NSE/BSE historical data; LLM-only tools cannot replicate this.
3. **Streaming architecture** – Pathway incremental indexing achieves <2s latency vs. static dashboards that recompute on every page load.
4. **Portfolio-aware alerts** – signals contextualised to each user's actual holdings, not generic screener results.

5. **Compound improvement** – each user interaction improves signal ranking; system accuracy scales with user base.

6.9 Assumptions (Stated Explicitly)

1. Win rates based on backtesting against Nifty 50 stocks with 3–5 year data
2. Average return estimates use median, not mean (to reduce outlier effect)
3. Time savings assumes basic digital literacy and daily market participation
4. Premium conversion rate of 0.3% is conservative (industry average is 0.5–1%)
5. Back-of-envelope calculations – detailed validation would require A/B testing
6. SOM (1.5L users) = 0.5% of SAM (3 Cr active traders) – conservative for a 5-year horizon
7. Revenue projections assume no price increase over 5 years (understates upside)
8. Cost reduction estimate assumes research content budget of ∼Rs. 20 Cr/year at ET Markets' scale
9. Year 3 and Year 5 user projections assume compounding growth of ∼100% and ∼150% respectively from Year 1 base